

Hurricane Disaster Response:

Relief System

Software Requirements Specification

ChadGPT:

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Table of Contents:

1. <i>Introduction</i>	3
2. <i>Stakeholders</i>	4
3. <i>Constraints</i>	7
4. <i>System Description</i>	7
5. <i>Business Use Cases</i>	8
6. <i>Functional Requirements</i>	9
7. <i>Non-Functional Requirements</i>	9
9. <i>Competitive Analysis</i>	10

Introduction

Purpose:

Since 1980, natural disasters have caused almost 3 trillion dollars worth of property damage; that's a little more than 60 billion dollars a year on average. Out of the overall outstanding damage that has been dealt by these storms collectively, approximately 1.5 trillion dollars is compromised from hurricane damage, which is over 50%, making hurricanes one of the most destructive and costly types of natural disasters in the United States.

In these types of disasters, people lose more than property and material assets; countless families and individuals lose access to basic necessities, like food, shelter, water, and medical aid to share a few. Support in these types of events is often widespread to the point of ineffectiveness; we plan to fix that issue.

Our project's goal is to ultimately develop a hurricane relief system that makes it easier for people to find and request the assistance they need before, during, or after a hurricane. The system will provide information about nearby shelters, medical assistance, food, water, and other available resources, while also allowing people to request or volunteer assistance. Bringing these resources together in one accessible platform will allow us to create an extremely efficient system that aims to reduce the difficulty of finding help during an already stressful and potentially life-threatening situation.

Scope:

In this document you will find:

- Stakeholders/ who will be using our app
- Constraints our the app
- A description of the functionality
- A competitive analysis to show what our app brings to the table that similar apps do not

Sources

[National Centers for Environmental Information](#)
[NOAA's Atlantic Oceanographic and Meteorological Laboratory](#)



Name: Bob Bufftin

Age: 28

Gender: Male

Occupation: Firefighter

Location: Coast Region

Firestation

Goals:

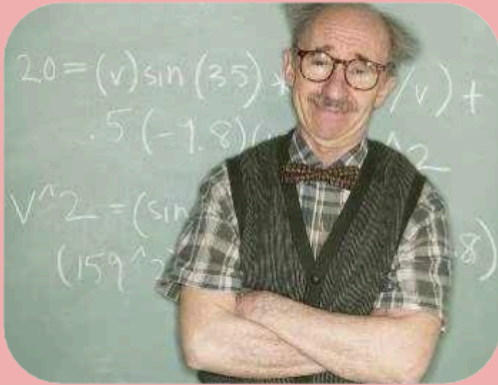
- Create a safer environment
- Help other people
- Support his family

Motivation: Growing up, Bob always had a kind heart, helping anyone who needed it. This led him to become a firefighter so he could help many people in his community.

Scenario 1: A hurricane warning is issued. Buff Bob finds residents who need transportation and takes them to shelters on his truck.

Scenario 2: A flood blocks a road near a neighborhood. Buff Bob checks for alternate routes to use.

Scenario 3: After the hurricane, a family reports being trapped in their home. Buff Bob coordinates with other volunteers to rescue the family.



Goals:

- Win a nobel prize
- Effectivley teach his students
- Become rich one day

Name: Dr. John Professor

Age: 68

Gender: Male

Occupation: Professor

Location: Coastal College

Town

Motivation: Dr. John has always been a brainiac, eager to attain more knowledge, but he also enjoys spreading said knowledge. For this, being a university proffessor was a no-brainer to him.

Scenario 1: A hurricane is hitting and Doctor Professor needs to find a shelter.

Scenario 2: Doctor Professor got injured while fleeing the hurricane and needs to find medical help near him.

Scenario 3: A tree fell and blocked Doctor Professor's driveway, and he cannot leave. He needs the help of a volunteer.



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Name: Samuel Swift

Age: 48

Gender: Male

Occupation: Emergency

Dispatcher

Location: Northeast Florida

Goals:

- Protect the city
- Save other people
- Make his dad proud

Motivation: At 12 years old, Samuel lost his father who was a cop. From that point forward Samuel decided he was going to make his father proud by becoming the best cop there ever was.

Scenario 1: Shelter A is getting close to maximum capacity, so Speedy Samwell changes the primary shelter to one of lesser fullness.

Scenario 2: A user reports a flooded road that isn't showing on major maps; Speedy Samwell pushes an alert letting people know to avoid said road.

Scenario 3: Our App/Website goes down; Speedy Samwell in the meantime helps users by providing phone support, and connecting with volunteers (by phone or radio to get work done).

Constraints

Monetary Constraints - There is no budget for this project.

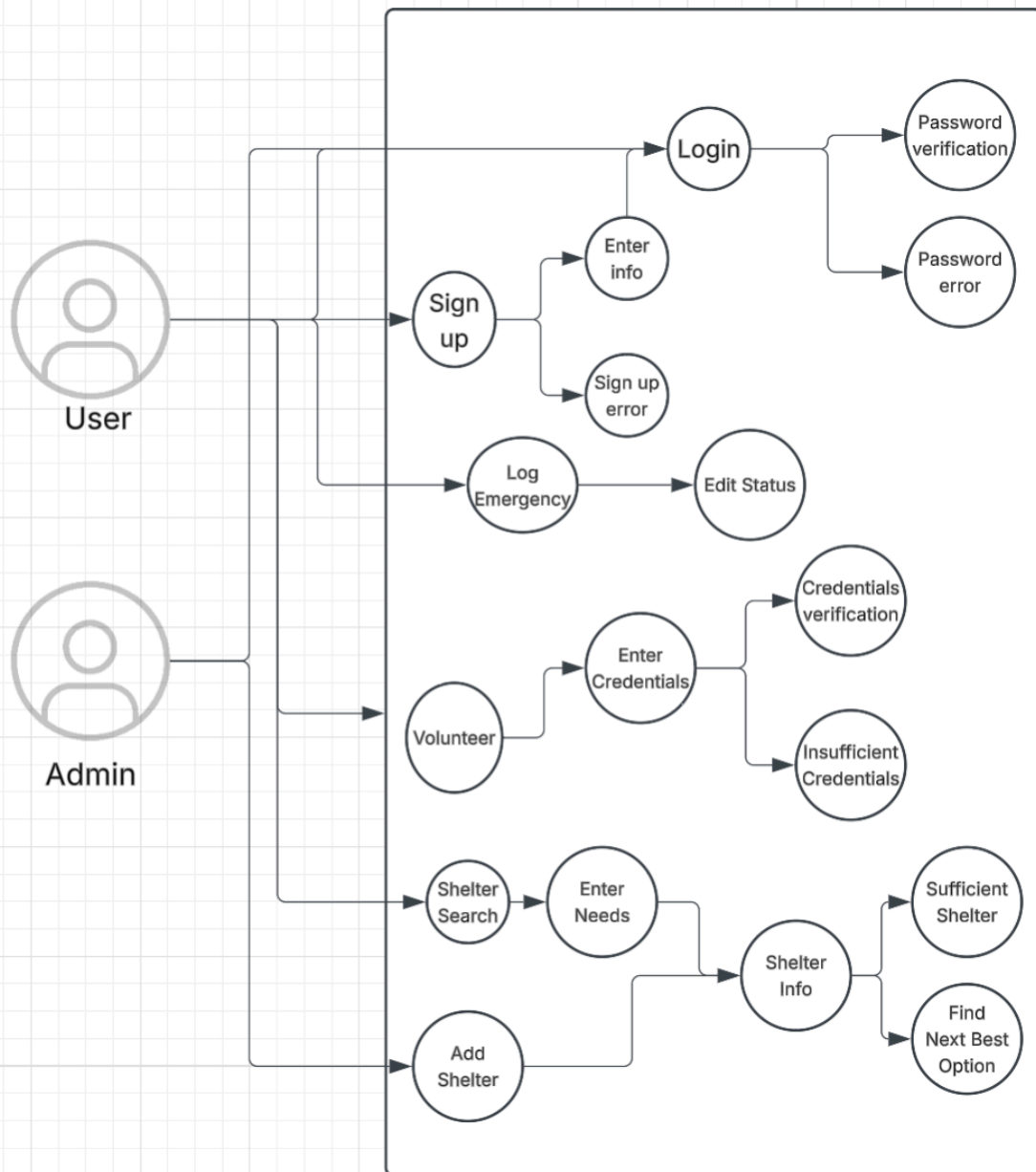
Technical Constraints - We have to use Java to create this website, and it needs to be available cross platform in order for people in need to easily access it no matter where they are.

Time Constraint - We have less than a semester to complete this project.

Overall Description

This hurricane relief app needs to be accessible to people who need help either recovering from a hurricane, during a hurricane, or before a hurricane. It should be accessible from mobile devices, laptops, or personal computers. The website will offer information on shelters nearby, where to get medical attention, food, or water, and will also include a portion dedicated to volunteering assistance or requesting assistance. It will include any source of help in the immediate area so that users may find help quickly, and through any source.

Business Use Cases:



Functional Requirements

<https://docs.google.com/spreadsheets/d/14SPIZaty4TjSweXMOOnU2cqUUw0YDPY3j2QQ4rCvAnk0/edit?usp=sharing>

Non-Functional Requirements

https://docs.google.com/spreadsheets/d/1THCIbXvix_S09hXR92zeJQ_Pi-YAi6Cwq7w4Xn9FXoU/edit?usp=sharing

Competitive Analysis

Salvation Army	
Strengths	A very appealing website to the eyes. The layout is very nice, and everything is neatly organized so that it is easy to find. It offers very many services to assist people, including disaster relief. They also make it very easy to find what you are looking for, as they have filters that you can apply when you search for assistance around you. There are also immediate prompts when you go onto the website which ask if you would like to volunteer and help, making it easy to get involved.
Weaknesses	The search is very slow when you're trying to find assistance, and it doesn't look like it encourages much of a community. All of the services are through the Salvation Army, and for some people that might not be the best/closest option when they're in a disaster scenario.
Audience/Focus	The website was created for people who need help. That could be homeless shelters, disaster relief, or even human trafficking cases.

Fema	
Strengths	The FEMA website has a lot of good information for people that are going through disasters. You can find shelters and recovery centers, apply for assistance, check disaster information, and then see if you are in the disaster area to qualify. FEMA also has a mobile app that gives alerts and helps people find nearby resources. Another strength is that FEMA has a lot of resources and funding ready to go, so they can help on a much larger scale than smaller organizations. They also work with local and state agencies to help coordinate disaster recovery and can prepare before some disasters happen by moving resources into areas that may need them.

Fema	
Weaknesses	The website can be too much because there is a lot of information spread across the different pages. Some of the wording is also very government based, so it may take longer for someone to figure out exactly where they need to go or what they need to do. FEMA is also not usually the first response agency, meaning they really just coordinate and provide resources rather than being the first ones going in and helping people directly. Because assistance can depend on requests from local and state agencies and certain eligibility requirements, (like area of disaster) getting federal help may also take time.
Audience/Focus	FEMA is mainly focused on people and communities affected by disasters. They also work with local and state governments, emergency responders, and other organizations to help coordinate disaster response and recovery.

Team Rubicon	
Strengths	The Team Rubicon website has a modern and visually interesting design that is easy to navigate and has clear organization. It has obvious buttons for volunteering and donating. The site also gives updates about current disaster responses, which help visitors see where help is needed. It focuses on using trained volunteers to help communities recover after hurricanes, floods, fires, and other disaster. Offering free resources to help train volunteers properly for disaster relief.
Weaknesses	The website mainly focuses on getting people to volunteer or donate. It does not make it as easy for a person affected by a hurricane to quickly find other resources and shelters after a hurricane. Some information about what Team Rubicon does is spread across different pages, which can make it confusing for older audiences navigating the website.
Audience/Focus	Team Rubicon's website is focused on people who want to volunteer for disaster relief, especially people interested in being properly trained and prepared to help with cleanup, recovery, and emergency response. It serves communities affected by disasters with the help of donors and other organizations to provide support and relief efforts.

Summary			
	Strengths	Weaknesses	Focus
Team Rubicon	+Organized scaffolding layout +Easy volunteer sign-up +Current disaster updates +strong community focus +Helps disaster cleanup and recovery	-Not focused on immediate emergency relief - No shelter or medical locator -Can be confusing the navigate multiple pages for resources	People who want to volunteer for disaster relief and/or donate to support communities recovering from hurricanes and other disasters.
Salvation Army	+Layout +Search System Filters +Volunteering	-No help finding -Slow search	People in danger/in need of help.
Fema	+Huge amount of resources/funding +Shelter/Recovery center locator +Mobile app and alerts +Works with the local/state agencies	-Not a first response agency -Federal aid takes time - Assistance depends eligibility (area of impact) - Complex application -Website is overwhelming	For people who are affected by disasters

Comparing all of these great websites we can find many strengths that they have, as well as some shortcomings that our team can learn from, and build a more versatile web-app. Team Rubicon had a strong application that streamlined volunteering and disaster cleanup, keeping a clean and organized look for their website. But they did fall short on their lack of help for critical emergencies. Salvation Army kept a polished layout and implemented a useful system search filter feature. While they did have a functional and useful new search feature, this came with long loading times for searches. Fema also had a great website that offered a large amount of resources for users to utilize like alerts, shelter locators,

and government aid. With this amount of resources came a complex website that could trouble some users who aren't very tech-savvy. We want to make an app that combines all of these websites' strengths, like resource abundance, and flips their weaknesses, like overly complex designs, while also adding some of our own unique features. We hope to create an app that streamlines the process of hurricane relief for all who are affected in a timely manner, creating safety and peace along the coasts.